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Curb parking occupancy prediction based on real-time fusion of multi-view spatial-temporal information using graph attention gated networks

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HIGHLIGHTS

- MGA-GRU was proposed for real-time curb parking occupancy prediction.
- MGA-GRU concurrently fused spatial-temporal data with graph attention gated units.
- Multi-view graph integrated geographic, accessibility, and POI data.
- MGA-GRU outperformed the baseline in forecasting curb parking occupancy.
- Our model provided valuable insights for improving parking system functionality.

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ABSTRACT

Effective curb parking management is crucial for reducing traffic congestion, minimizing cruising time, and lowering pollution in smart cities through accurate, timely occupancy predictions. However, traditional spatial-temporal fusion methods often rely on sequential concatenation, leading to spatial information lag and limited real-time fusion. These methods also overlook critical spatial features for curb parking, such as accessibility metrics. To address these limitations, this study introduces the Multi-View Graph Attention Gated Recurrent Unit (MGA-GRU) model, which innovatively fuses spatial-temporal data via graph attention gated networks to enhance curb parking occupancy predictions. Validated on a curb parking dataset from Lanzhou city, China (July 2019 to December 2020), the MGA-GRU model significantly outperforms baseline methods in both short-term and long-term predictions. Ablation experiments demonstrate that each component—the graph attention gates, multi-view graph structure, and real-time fusion—significantly enhances the model's predictive accuracy, highlighting their collective importance in advancing occupancy prediction. By refining spatial-temporal fusion unit, the MGA-GRU model offers vital insights for optimizing curb parking resources and supports efficient traffic management and sustainable urban development.

1. Introduction

Curb parking poses a significant challenge in urban areas globally due to population growth, rising vehicle numbers and sizes, limited parking availability, and parking lot locations. The surge in vehicle ownership has worsened the struggle to secure adequate parking spaces within urban transportation systems [1,2]. This has led to issues such as cruising, parking violations, and regional traffic congestion [3]. According to Cookson & Pishue [4], New York City drivers spend an average of 107 hours annually searching for parking spaces, resulting in significant financial burdens and exacerbating congestion issues. This statistic highlights not only the frustration and time wastage experienced by drivers but also the economic toll of inefficient parking systems, including expenses related to wasted time, fuel consumption, and emissions treatment. A similar situation is observed in Lanzhou, a city in Northwest China, where vehicles spend prolonged periods searching for curb parking spaces, exacerbating road congestion. Furthermore, the decision to opt for private vehicles over public transportation, driven by

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individual preferences such as enhanced comfort and convenience, exacerbates the problem [5]. The escalating number of vehicles, coupled with limited curb parking availability and geographical constraints, exerts significant implications on traffic congestion and environmental sustainability, particularly during peak hours and in densely populated commercial areas [6]. Empirical evidence from cities like New York illustrates the magnitude of these issues, with drivers spending considerable time and resources dealing with the complexities of urban curb parking. While current curb parking management systems offer real-time parking occupancy information, they lack the ability to predict future parking occupancy. Therefore, with the growing number of vehicles and limited urban space, there is increasing interest in curb parking occupancy prediction. This task is crucial for modern curb parking management systems as it demonstrates their intelligence and can lead to increased economic benefits.

Various techniques have been developed for the parking occupancy prediction, including traditional temporal models like ARIMA [7,8], DT [9], SVR models [10,11] (A glossary of abbreviations was presented in Table 1). However, parking occupancy data often exhibits long-term dependencies and complex nonlinear patterns, which traditional temporal models struggle to capture effectively. To tackle these limitation, advanced methods like TCN networks [12], LSTM networks [13–15] and GRU networks [16,17] have been applied with promising results. These deep learning models significantly improve predictive accuracy for curb parking occupancy by learning intricate temporal patterns that traditional models cannot process. Nevertheless, they often lack spatial awareness, limiting their effectiveness in predicting parking occupancy. To address this gap, scholars have introduced the GNNs to incorporate spatial features, which have demonstrated promising results in modeling the spatial dependencies in curb parking occupancy [18–21]. GNNs leverage the topological structures and spatial relationships between parking locations, effectively capturing complex patterns and connections in irregular and non-uniform spatial data. However, most existing methods use a sequential approach, first applying GNNs to model spatial features, then using recurrent neural networks to process temporal features. This tandem fusion of spatial-temporal feature fusion can lead to spatial information lag and insufficient integration, ultimately compromising prediction accuracy. Our study addresses these challenges by proposing a novel approach that enables real-time fusion of spatial-temporal features in curb parking data. This integrated approach enhances prediction accuracy by capturing the intricate interdependencies between spatial and temporal aspects of parking occupancy. Moreover, existing methods typically rely on geographic distance alone to define spatial relationships [19], overlooking other crucial spatial factors such as accessibility and POI. To improve prediction reliability, our approach incorporates these additional spatial features, recognizing their importance in accurately forecasting curb parking occupancy.

Table 1
Glossary of abbreviations.

Abbreviations	Description	Abbreviations	Description
ARIMA	Autoregressive Integrated Moving Average	*MAE*	Mean Absolute Error
Acc	Accuracy	*MedAE*	Median-Absolute Error
CNN	Convolutional Neural Network	MGA-GRU	Multi-View Graph Attention Gated Recurrent Unit
CRNN	Convolutional Recurrent Network	*MSE*	Mean Squared Error
CPPM	Correlated Parking Prediction Model	*MGCN*	Multiple-Graph Convolutional Neural Network
DT	Decision Tree	NN	Neural Networks
GA	Graph Attention	OD	Origin-Destination
GAT	Graph Attention Network	POI	Points of Interest
GCN	Graph Convolutional Network	R^2	Coefficient of Determination
GCNRNN	Graph Convolutional Recurrent Neural Network	RT	Regression Tree
GNNs	Graph Neural Networks	RFR	Random Forest Regression
GRU	Gated Recurrent Unit	*RMSE*	Root Mean Squared Error
HAPSN	Historical Available Parking Space Number	RNN	Recurrent Neural Network
HPOR	Historical Parking Occupancy Rate	SARIMA	Seasonal Autoregressive Integrated Moving Average
HST-GCNs	Hybrid Spatial-Temporal Graph Convolution Networks	SVR	Support Vector Regression
FAPSN	Future Available Parking Space Number	TCN	Temporal Convolutional Network
FPOR	Future Parking Occupancy Rate	T-GCN	Temporal Graph Convolutional Network
LSTM	Long Short-Term Memory	*var*	Explained Variance Score
LTDCG	Lanzhou Traffic Development and Construction Group	XGBoost	Extreme Gradient Boosting

Despite advancements in curb parking occupancy prediction, significant challenges remain in the effective real-time fusion of spatial and temporal features, which is critical for accurate predictions. Traditional models often rely solely on geographic proximity, failing to incorporate other key spatial factors such as accessibility and POI, which are essential for reflecting true curb parking dynamics. To address these issues, we propose the Multi-View Graph Attention Gated Recurrent Network model, which incorporates diverse spatial views and employs a graph attention mechanism within a gated recurrent structure. This model enables real-time, comprehensive spatial-temporal fusion, offering a robust solution to the limitations of prior approaches and significantly enhancing curb parking occupancy prediction accuracy.

Major contributions of this paper are outlined as below:

(1) **Innovative architecture:** Our investigation introduces the MGA-GRU architecture specifically designed for curb parking occupancy prediction. This architecture distinguishes itself by enabling the seamless real-time fusion of spatiotemporal data through the integration of dual graph attention gates within the GRU network.
(2) **Enhanced spatial characterization:** In our study, we introduce a multi-view graph that integrates geographic distance, accessibility, and POI data to enhance the precision and comprehensiveness of spatial characteristics in the context of curb parking prediction. By amalgamating these diverse sources of information, our approach aims to provide a holistic understanding of the spatial dynamics influencing curb parking occupancy.
(3) **Case study and performance evaluation:** The effectiveness of the proposed method is validated through a case study conducted in the central area of Lanzhou city. The results demonstrate that the MGA-GRU model significantly outperforms existing baseline methods, thereby advancing the functional capabilities of curb parking systems by delivering more accurate predictions of future parking availability.

The rest of this paper is organized as follows. Section II presents related work summarizing forecasting models and their applications in curb parking occupancy prediction. Section III describes the study methods, including the overview, problem definition, and the fusion modeling algorithm. Section IV provides a case study encompassing data description, evaluation metrics, benchmark models, model parameters, results, ablation study, and discussion. Finally, Section V concludes the study by summarizing the key findings and contributions to curb parking occupancy prediction, and highlights potential directions for future research.

2. Related work

In this section, related models for temporal-based and spatial-temporal-based in the parking occupancy prediction task are summarized.

2.1. Temporal-based models

The accurate prediction of parking occupancy depends on various temporal factors, encompassing traffic flow, weather conditions, and holidays. Time-based models, such as ARIMA, SVR, and DT, effectively capture the periodicity and interrelationships among these temporal features, thereby enabling precise predictions [7–11,22]. For instance, the ARIMA model was employed for parking occupancy prediction and compared its performance with neural networks [7]. In order to capture nonlinear relationships more effectively, experiments were conducted on two datasets by using SVR, regression trees, and neural networks [10]. Moreover, the DT model was utilized to forecast the number of available parking spaces by tuning the hyperparameter max depth and the DT model outperformed the RFR in term of *MSE*, *MAE*, *RMSE*, and R^2 metrics [9]. However, it is important to note that machine learning methods have limitations in feature engineering and handling nonlinear relationships, which may constrain the model's performance in complex tasks and large-scale data scenarios.

In recent years, deep learning methods have been widely applied in the parking field due to their ability to extract discriminative features and capture complex correlations [23]. Therefore, several deep learning models have been employed for parking occupancy prediction task, such as TCN networks [12], LSTM networks [13–15] and GRU networks [16,17]. As an illustration, the convolution and dilation of one-dimensional convolutional network architecture through TCN was utilized to dynamically adaptive to the prediction window and prediction response of parking occupancy prediction [12]. Additionally, the effectiveness of combining XGBoost and LSTM was demonstrated in predicting truck occupancy [14]. Similarly, scholars devised a novel hybrid model that integrated GRU and LSTM architectures, alongside occupancy data, weather conditions, and holiday information, aiming to improve the precision of parking occupancy forecasts [16]. While traditional machine learning approaches may be less complex, their prediction accuracy and reliability often fall short. In contrast, deep learning methods offer superior nonlinear fitting capabilities and higher prediction accuracy, making them a preferable choice over traditional machine learning approaches.

2.2. Spatial-temporal-based models

Different from the aforementioned temporal-based models, various spatial variables, including geographical proximity and land utilization, have been used for the parking occupancy prediction task in the spatial-temporal models. The crux of the spatial-temporal model lies in the fusion of spatial and temporal information, where the spatial model extracts spatial features and the temporal model extracts temporal features. Several spatial-temporal models have been developed for parking occupancy prediction task, including the CRNN [24,25] and GCNRNN [18–21,26]. For instance, scholars proposed the Hybrid-PLO model, which integrated CNN and LSTM methods for parking occupancy prediction task [25]. Besides, the HST-GCNs model was proposed to obtain spatial and temporal features using graph convolutional networks and gated linear units with CNN [19]. Moreover, the residual spatial-temporal graph convolutional neural network was utilized to capture and fuse the spatial and temporal features through GNN, TCN and residual spatial-temporal convolutional block [21]. The methods above have achieved significant progress in the parking occupancy prediction task by incorporating CNN and GCN networks to capture spatial dependencies. However, they rely on tandem concatenation in spatial and temporal feature fusion, extracting spatial and temporal features sequentially, which impedes achieving real-time fusion of spatial-temporal features. Moreover, CNN is less suitable for graph-based spatial feature extraction. Although GCN can process graph data, it lacks the ability to adaptively learn the importance of relationships between nodes, leading to inaccurate capture of global information.

Besides, selecting spatial information is also the key to the curb parking occupancy prediction task. Currently, most studies choose geographic location to represent spatial information, but some scholars have also explored alternative spatial information, such as POI. For example, scholars comprehensively mined contextual relationships between nodes by constructing four different graphs [27]. Furthermore, to investigate the impact of geospatial information on parking occupancy prediction models, scholars employed random forest and artificial neural network techniques to integrate road network centrality, land use, and POI for the parking occupancy prediction task [28].

Table 2 presents an overview of prior studies focused on curb parking occupancy prediction in various areas. Each study was categorized by area, input and output data, predictive models employed, evaluation metrics, and references. For example, in Nanjing, China, the HASPN and FASPN were modeled using ARIMA, with metrics such as *MAE*, *RMSE*, and *MAPE* used for evaluation. This compilation provides insights into methodological approaches and performance metrics across different studies in the field.

In light of the context above, our study introduces a new neural network architecture to addresses two key issues. Firstly, it enables the real-time fusion of spatial-temporal information by incorporating two GA gates into the GRU unit. Secondly, it constructs an integrated graph to aggregate multi-view spatial information of curb parking.

3. Methods

3.1. Overview

In this section, we delineate the utilization of the MGA-GRU model for curb parking occupancy prediction. Specifically, the proposed model begins by extracting temporal features from the curb parking occupancy data using the update and reset gates within the MGA-GRU unit, as depicted in Fig. 1. Furthermore, to extract spatial features, a multi-view spatial graph ($\mathbf{G}$) is formed by fusing the various spatial graphs of curb parking occupancy prediction ($\mathbf{G}_1,\ldots,\mathbf{G}_K$). Subsequently, GA_1 and GA_2, denoting the initial and secondary graph attention gates respectively, are employed to merge the candidate hidden feature with spatial information and the existing hidden feature with spatial information, thereby generating the ultimate hidden state for the current unit. Ultimately, the output from the last unit undergoes processing via a fully-connected layer to produce the final prediction result of the model. The GA modules are integrated into the gated recurrent unit as gates to facilitate real-time fusion of multi-view spatial-temporal information. The GA modules effectively capture the distinct relationships between nodes in the graph structure, while the gated recurrent unit handles long-term dependencies in temporal data.

Accordingly, the curb parking occupancy prediction task can be conceptualized as the process of training a parameterized model f_θ to forecast the future curb parking occupancy ($\widehat{\mathbf{x}}_{t+1},\ldots,\widehat{\mathbf{x}}_{t+F}$), leveraging the various graphs ($\mathbf{G}_1,\ldots,\mathbf{G}_K$) and historical curb parking occupancy ($\mathbf{x}_{t-H+1},\ldots,\mathbf{x}_{t-1},\mathbf{x}_t$) of N parking lots, as depicted in Eq. (1),

$$[\widehat{\mathbf{x}}_{t+1},\ldots,\widehat{\mathbf{x}}_{t+F}] = f_\theta((\mathbf{G}_1,\ldots,\mathbf{G}_K);(\mathbf{x}_{t-H+1},\ldots,\mathbf{x}_{t-1},\mathbf{x}_t)) \quad (1)$$

where F is the predicted time length, t is the current time, H is the

Table 2
Previous studies on curb parking prediction across areas.

Areas/Countries	Input	Output	Model	Metrics	References
Nanjing/China	HASPN	FASPN	ARIMA	*MAE*, *RMSE*, *MAPE*	[7]
San Francisco/USA	HPOR, Day of week, Time of day	FPOR	ARIMA, NN, SARIMA	*MSE*	[8]
Kuala Lumpur/ Malaysia	HASPN	FASPN	DT	R^2, *MSE*, *MAE*, *RMSE*	[9]
San Francisco/USA, Melbourne/Australia	Time of day, Day of week, HPOR	FPOR	RT, SVR, NN	R^2, *MSE*, *MAE*	[10]
Hangzhou/China	HASPN	FASPN	FT-LSSVR	R^2, *MSE*, *MAE*, *MRE*	[11]
Melbourne/Australia	HASPN	FASPN	TCN	*MSE*, *MAE*, *RMSE*	[12]
Melbourne/Australia	HASPN, Temperature, Pressure, Humidity, Wind speed	FASPN	LSTM	*MAE*, *RMSE*	[13]
Bavaria/German	HPOR, Month, Weekday, Hour, Minute	FPOR	XGBoost-LSTM	*RMSE*, *MSE*, *MAE*, *MedAE*	[14]
Chongqing/China	HPOR, Weather conditions, Holiday	FPOR	GRU-LSTM	*RMSE*, *MSE*, *MAE*	[15]
Istanbul/Turkey	Date-time, capacity, empty capacity, Location density rate, Average air temperature, Total amount of precipitation, Time of Day, Day of week, Public holiday, HPOR	FPOR	GRU	*RMSE*, *MSE*, *MAE*	[16]
Antwerp/Belgium, Barcelona/ Spain, Wattens/Austria, Los Angeles/USA	Day of week, Season, Weekday and holiday, Temperature, Wind speed, Precipitation intensity, Weather summary, HPOR	FPOR	GRU, LSTM	*RMSE*	[17]
Ningbo, Beijing/ China	HPOR, Spatial graphs	FPOR	CPPM	*RMSE*, *MSE*, *Acc*, *var*, R^2	[18]
Melbourne/Australia	HPOR, Spatial graphs	FPOR	HST-GCNs	*RMSE*, *MAE*, *MAPE*	[19]
Shenzhen/China	HASPN number, Time, Road network, POI, Weather, Holiday	FASPN	MGCN+LSTM	*RMSE*, *MAE*	[20]

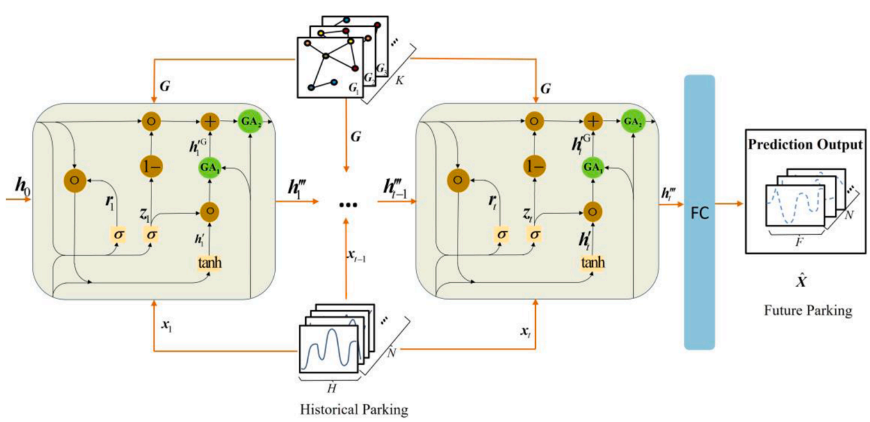

Fig. 1. The network structure of MGA-GRU model.

historical time length, K is the number of graphs.

3.2. Problem definition

We can construct various graphs to integrate diverse spatial factors in accordance with the requirements of curb parking occupancy prediction research. In this section, we define the graphs based on geographical distance, accessibility, and POI, with the value of K in Eq. (1) set to 3.

Definition 1. Graph based on geographical distance ($\boldsymbol{G}_1$). Geographical distance is commonly utilized in constructing graphs for curb parking occupancy prediction. The haversine formula calculates geographical distances using latitude and longitude converted into the radian system ($rlat$, $rlon$), as illustrated in Eq. (2),

$$h^D(v_i, v_j) = 2arcsin\sqrt{\sin^2\left(\frac{rlat_i - rlat_j}{2}\right) + \cos(rlat_i)\cos(rlat_j)\sin^2\left(\frac{rlon_i - rlon_j}{2}\right)} \tag{2}$$

where $rlat$ and $rlon$ represent the latitude and longitude of the curb parking lot in the radian system, $h^D(v_i, v_j)$ represents the geographical distance between parking lots v_i and v_j.

After calculating $h^D(v_i, v_j)$, a threshold D is selected to determine whether the lots are connected based on Eq. (3),

$$\alpha_{ij}^{D} = \begin{cases} 1 & \text{if} h^{D}(v_i, v_j) < D \\ 0 & \text{else} \end{cases} \tag{3}$$

where α_{ij}^{D} represents the interaction score based on the geographical distance between v_i and v_j, if $h^{D}(v_i, v_j) < D$, α_{ij}^{D} is 1, otherwise, it is 0.

Definition 2. Graph based on accessibility ($\mathbf{G}_2$). In addition to geographical distance, accessibility is also one of the crucial factors influencing curb parking occupancy. Scholars have noted a significant correlation in parking occupancy among adjacent parking lots [29]. In situations where parking lots reach full capacity, the accessibility of each lot becomes a pivotal spatial factor in determining occupancy. Consequently, parking lots with better accessibility will experience an increase in occupancy from their initial state. Considering the varying levels of accessibility is crucial when addressing the curb parking occupancy prediction task.

The OD cost matrix method [30] is commonly utilized to assess accessibility on road networks, constructed based on the road network itself. The matrix elements represent the shortest road lengths between the connected lots v_i and $v_j$denoted as $l^{R}(v_i, v_j)$.The time $t^{R}(v_i, v_j)$ from lot v_i to lot v_j is calculated using Eq. (4),

$$t^{R}(v_i, v_j) = \frac{l^{R}(v_i, v_j)}{v^{R}} \tag{4}$$

where v^{R} represents the vehicle travel speed around the curb parking lot, set at 20 km/h considering the actual road congestion situation in Lanzhou city and referencing relevant literature [31]. The accessibility between lots is measured using the arrival time $t^{R}(v_i, v_j)$, where shorter arrival time indicates better accessibility between adjacent parking lots and stronger the interaction among them. After calculating the accessibility of all curb parking lots, the accessibility graph is constructed by selecting the threshold R to determine connectivity between lots, based on Eq. (5),

$$\alpha_{ij}^{R} = \begin{cases} 1 & \text{if} t^{R}(v_i, v_j) < R \\ 0 & \text{else} \end{cases} \tag{5}$$

where α_{ij}^{R} represents the interaction score based on accessibility between v_i and v_j, if $t^{R}(v_i, v_j) < R$, α_{ij}^{R} is 1, otherwise, it is 0.

Definition 3. Graph based on POI ($\mathbf{G}_3$). In the context of predicting curb parking occupancy, the distribution and quantity of POI serve as crucial spatial factors. A higher concentration of POI near a parking lot correlates with increased parking occupancy rates for that facility. Therefore, when constructing a spatial graph, it's essential to consider both the quantity and spatial arrangement of POI surrounding a curb parking lot. Consequently, our study primarily focuses on several major public POI, including hotels, tourist attractions, educational institutions, local markets, shopping centers, medical facilities, and office complexes. To assess the influence of POI distribution on the interconnectivity between parking lots, a 500-meter radius area centered on each curb parking lot is defined based on existing literature [32]. The normalized count of POI in the overlapping area of two parking lots quantitatively represents their level of interaction, as indicated in Eq. (6),

$$n^{P} = \frac{\exp\left(N_{poi \in P_{ij}}\right)}{\exp\left(N_{poi}\right)} \tag{6}$$

where n^{P} represents the normalized count of major public POI within the shared area, $N_{poi \in P_{ij}}$ is the count of major public POI within the common area (P_{ij}), P_{ij} denotes the common overlap area between parking lots v_i and v_j, and N_{poi} signifies the total count of major public POI across all areas.

The construction of the POI graph involves setting a threshold B to assess the level of interaction between parking lots, as depicted in Eq. (7),

$$\alpha_{ij}^{P} = \begin{cases} 1 & \text{if} n^{P} > B \\ 0 & \text{else} \end{cases} \tag{7}$$

where α_{ij}^{P} represents the interaction score between v_i and v_j, if $n^{P} > B$, α_{ij}^{P} is 1, otherwise, it is 0.

3.3. Model

3.3.1. Spatial features modeling

Spatial feature modeling constitutes a pivotal stage in predicting curb parking occupancy. GCN [18] and GAT [33] stands as popular methods for modeling spatial features with graph in deep learning. Among these, GAT stands out for exceptional capacity to capture relationships, node importance, flexibility, interpretability, handling large graph data, and robust generalization ability. Consequently, this study utilizes the graph attention mechanism for spatial feature modeling, wherein neighboring nodes features are aggregated based on varying feature weights.

Firstly, to facilitate expression, the multi-view interaction scores of each parking lot in Section 3.2 are transformed into a three-dimensional vector $\boldsymbol{d}_{ij} = [\alpha_{ij}^{D}, \alpha_{ij}^{R}, \alpha_{ij}^{P}]^{T}$, and this vector is utilized to calculate the final spatial relation between parking lot v_i and v_j according to Eq. (8),

$$\alpha_{ij} = \frac{\exp\left(\boldsymbol{d}_{ij}{}^{T}\boldsymbol{e}\right)}{\sum_{j=1}^{N_i}\exp\left(\boldsymbol{d}_{ij}{}^{T}\boldsymbol{e}\right)} \tag{8}$$

where $\boldsymbol{e}$ is a 3-dimensional unit column vector, α_{ij} is the element of spatial graph ($\boldsymbol{G}$). Secondly, graph attention networks are employed to aggregate spatial information from multiple sources, addressing limitations posed by CNN on non-Euclidean graphs and the shortcomings of GCN in non-differential aggregation of neighbor node information. The specific implementation process is as follows:

$$e_{ij} = \mathrm{LeakyReLU}\left(\boldsymbol{\theta}^{\mathrm{T}}\alpha_{ij}\left[\boldsymbol{W}\boldsymbol{h}_i \middle|\middle| \boldsymbol{W}\boldsymbol{h}_j\right]\right) \tag{9}$$

$$\beta_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in N_i}\exp(e_{ik})} \tag{10}$$

$$\boldsymbol{h}_i^{G} = GA(\mathbf{h}) = \sigma\left(\sum_{j \in N_i}\beta_{ij}\boldsymbol{W}\mathbf{h}_j\right) \tag{11}$$

where $\boldsymbol{\theta}$ and $\boldsymbol{W}$are the parameters to be learned, $\boldsymbol{h}_i$ and $\boldsymbol{h}_j$ represent the historical hidden feature of parking lot v_i and v_j respectively, the symbol $||$ denotes the vector concatenation operation, N_i denotes all neighboring nodes (including v_i) of parking lot v_i, $\boldsymbol{h}$ represents the historical hidden feature of N_i, $\boldsymbol{h}_i^{G}$ is the graph hidden feature that integrates spatial graph feature. Eq. (9) calculates the correlation (e_{ij}) between lots, with larger values of e_{ij} indicating a more significant influence of spatial information, the attention weights β_{ij} between v_i and v_j are calculated using Eq. (10). Subsequently, the GA gate is utilized to fuse the temporal hidden feature and spatial graph feature according to Eq. (11). The GA modules effectively capture the relationship and importance among curb parking lots in spatial data. It focuses on significant lots and edges through an

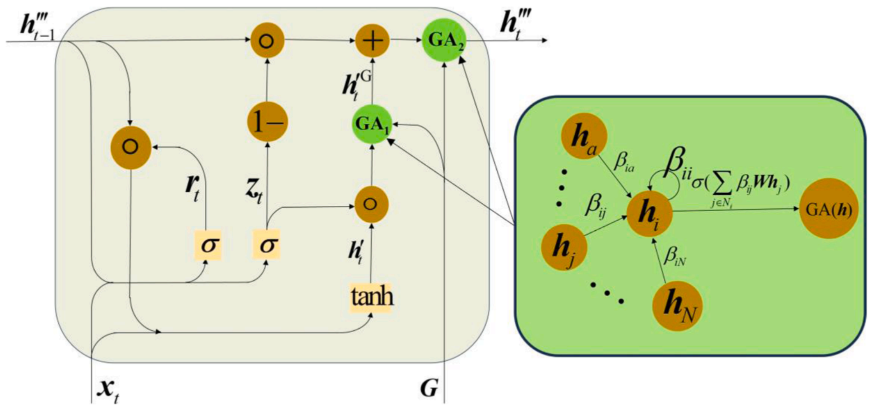

Fig. 2. The MGA-GRU unit structure.

attention mechanism, thereby enhancing the ability of modeling spatial features.

3.3.2. Temporal features modeling

In predicting curb parking occupancy, temporal feature modeling is essential, with deep learning methods being extensively utilized. RNNs [34,35] and TCNs [36] have gained widespread adoption. RNNs, particularly, are prevalent due to their recurrent structure and adeptness at capturing long-term dependencies, rendering them suitable for variable-length sequence data processing. Their interpretability and parameter sharing further enhance their utility in modeling temporal features for curb parking occupancy prediction. Variants like LSTM [37] and GRU [38] integrate gated mechanisms to retain long-term information, addressing gradient disappearance and explosion. In this study, the GRU model, with its simpler structure, fewer parameters, and faster training capabilities, is chosen to effectively capture temporal dependencies. The GRU model generates the prediction result by combining the input at the current moment with the hidden state through the update gate and reset gate. The information transfer process is represented by the following equations:

$$\mathbf{z}_t = \mathrm{Sigmoid}(\mathbf{W}_z \boldsymbol{x}_t + \boldsymbol{U}_z \boldsymbol{h}_{t-1} + \boldsymbol{b}_z) \tag{12}$$

$$\boldsymbol{r}_t = \mathrm{Sigmoid}(\mathbf{W}_r \boldsymbol{x}_t + \boldsymbol{U}_r \boldsymbol{h}_{t-1} + \boldsymbol{b}_r) \tag{13}$$

$$\boldsymbol{h'}_t = \tanh(\mathbf{W}_{h'} \boldsymbol{x}_t + \boldsymbol{U}_{h'} (\boldsymbol{r}_t \circ \boldsymbol{h}_{t-1}) + \boldsymbol{b}_{h'}) \tag{14}$$

$$\boldsymbol{h}_t = (\mathbf{1} - \boldsymbol{z}_t) \circ \boldsymbol{h}_{t-1} + \boldsymbol{z}_t \circ \boldsymbol{h'}_t \tag{15}$$

where $\circ$ represents the Hadamard product of matrices; Sigmoid and tanh are the activation functions; $\boldsymbol{r}_t$, $\boldsymbol{z}_t$ and $\boldsymbol{h'}_t$ are the reset, update and new gates, respectively; $\boldsymbol{W}_z$, $\boldsymbol{U}_z$, $\boldsymbol{W}_r$, $\boldsymbol{U}_r$, $\mathbf{W}_{h'}$ and $\boldsymbol{U}_{h'}$ are the weight matrices; $\boldsymbol{b}_z$, $\boldsymbol{b}_r$ and $\boldsymbol{b}_{h'}$ are the bias terms; $\boldsymbol{x}_t$ represents the input at the current state; $\boldsymbol{h}_{t-1}$ stands for the hidden feature of the previous state; $\boldsymbol{h}_t$ denotes the hidden feature of the current state; $\boldsymbol{h'}_t$ represents the candidate hidden feature of the current state; $\boldsymbol{z}_t$ signifies the update gate controlling the passage of current state information into the current unit; $\boldsymbol{r}_t$ is the reset gate regulating the transmission of previous state information to $\boldsymbol{h'}_t$. The GRU model captures evolving patterns in historical data and establishes temporal correlations while obtaining current curb parking occupancy information.

3.3.3. Features fusion modeling

Spatial-temporal data are characterized by high dynamics and complexity, requiring rapid and accurate capture of key features and patterns. These data typically consist of multiple dimensions and types of features, necessitating their fusion and modeling to enhance prediction accuracy and reliability. Therefore, real-time spatial-temporal feature fusion modeling is crucial for accurate spatial-temporal predictions.

The MGA-GRU unit incorporates two graph attention gates (GA_1 and GA_2) to synchronously fuse the spatial-temporal information during data transmission. Fig. 2 details the specific structure of this unit. Initially, the unit receives the current curb parking occupancy ($\boldsymbol{x}_t$) and the previous hidden state ($\boldsymbol{h}_{t-1}$) to extract temporal feature via the update and reset gates, as defined based on Eq. (12) and Eq. (13). Subsequently, the multi-view spatial graph ($\boldsymbol{G}$) is introduced, and the attention weights (β_{ij}) between nodes and their neighbors are calculated and normalized using the softmax function, as illustrated in Eq. (9) and Eq. (10). Following this, GA_1 is integrated after the candidate hidden state output ($\boldsymbol{h'}_t$) to fuse the candidate hidden feature and spatial information according to Eq. (11) and Eq. (16). Then, the update gate updates the previous hidden features ($\boldsymbol{h}_{t-1}$) and the candidate hidden features fused with spatial features ($\boldsymbol{h'}_t{}^G$) based on Eq. (17). Ultimately, GA_2 is embedded after the hidden state output ($\boldsymbol{h}_t^{''}$) to fuse the hidden feature and spatial information, producing the final hidden state ($\boldsymbol{h}_t^{'''}$) of the gated unit according to Eq. (18). GA_1 and GA_2 perform the spatial-temporal information extraction and fusion concurrently, facilitating real-time fusion of spatial and temporal information throughout the data transmission process.

$$\boldsymbol{h'}_t{}^G = GA_1(\boldsymbol{h'}_t) \tag{16}$$

$$\boldsymbol{h}_t^{''} = \boldsymbol{z}_t \circ \boldsymbol{h'}_t{}^G + (\mathbf{1} - \boldsymbol{z}_t) \circ \boldsymbol{h}_{t-1} \tag{17}$$

$$\boldsymbol{h}_t^{'''} = GA_2\left(\boldsymbol{h}_t^{''}\right) \tag{18}$$

3.3.4. Loss function

In the training process, the training objective of the MGA-GRU model aims to minimize the error between the actual values $\boldsymbol{X}$ and the predicted values $\widehat{\boldsymbol{X}}$ of samples. Therefore, the loss function of the MGA-GRU model is defined as:

$$loss = \|\boldsymbol{X} - \widehat{\boldsymbol{X}}\|^2 + \lambda \boldsymbol{L}_{reg} \tag{19}$$

The first term in Eq. (19) minimizes the error between the actual sample values and the predicted values, and the second term serves as a regularization term that helps mitigate overfitting issues, where λ is a hyperparameter.

3.3.5. Pseudocode for the MGA-GRU model

Algorithm 1. The framework of MGA-GRU

Input: $\boldsymbol{X}$: the historical parking occupancy data
epoch: the maximum number of model training epochs
iter: the maximum number of iterations per epoch
$\boldsymbol{G}$: the multi-view spatial graph stacked by batch size
$\boldsymbol{h}_{t-1}$: the previous hidden state

Output: the trained model parameter

1: function MINIMIZE ($loss = \left\| \boldsymbol{X} - \hat{\boldsymbol{X}} \right\|^2 + \lambda \boldsymbol{L}_{reg}$)
2: **for** $i = 1 : epoch$ **do**
3: **for** $j = 1 : iter$ **do**
4: Calculate the update ($\boldsymbol{z}_t$), reset ($\boldsymbol{r}_t$) and candidate hidden ($\boldsymbol{h}'_t$) features based on Eq. (12) and Eq. (13).
5: Calculate and normalize the attention weights (β_{ij}) between nodes and their neighbors based on Eq. (9) and Eq. (10).
6: Calculated the fused hidden features ($\boldsymbol{h}'^{\mathrm{G}}_{\mathrm{t}}$) using the candidate hidden features ($\boldsymbol{h}'_t$) and spatial features (β_{ij}) based on Eq. (11) and Eq. (16).
7: Update the hidden features ($\boldsymbol{h}''_{\mathrm{t}}$) using the previous hidden features ($\boldsymbol{h}_{t-1}$) and the fused hidden features ($\boldsymbol{h}'^{\mathrm{G}}_{\mathrm{t}}$) based on Eq. (17).
8: Fuse the updated hidden features ($\boldsymbol{h}''_{\mathrm{t}}$) and spatial features (β_{ij}) based on Eq. (18).
9: Produce the model's prediction by a fully-connected layer.
10: Calculate the loss based on Eq. (19).
11: Update the model parameter.
12: **end for**
13: **end for**
14: **return** trained model parameter
15: **end** function

The specific training process of the MGA-GRU model is shown in Algorithm 1. Throughout the forward propagation process, each MGA-GRU unit realizes the extraction and real-time fusion of spatial-temporal features. GA_1 and GA_2 combine the candidate hidden

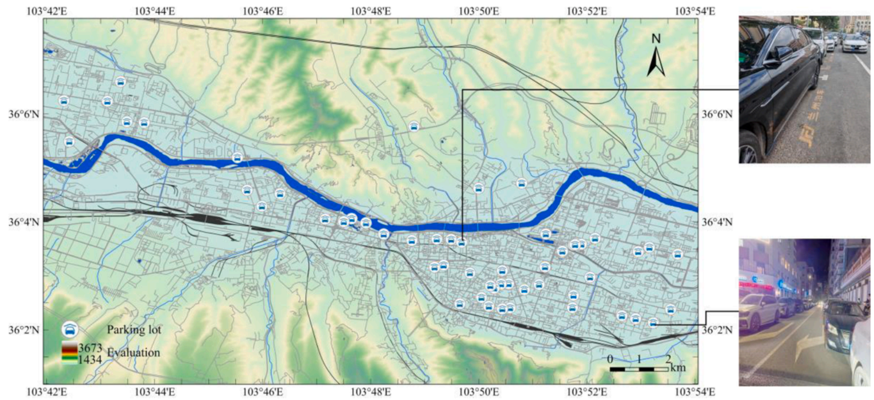

Fig. 3. Curb parking lots distribution in the urban district of Lanzhou.

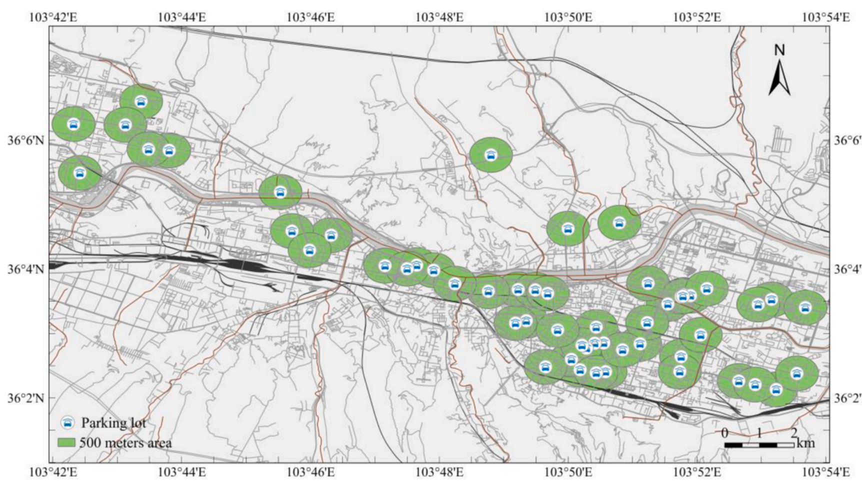

Fig. 4. Spatial graph based on geographical distance in the urban district of Lanzhou.

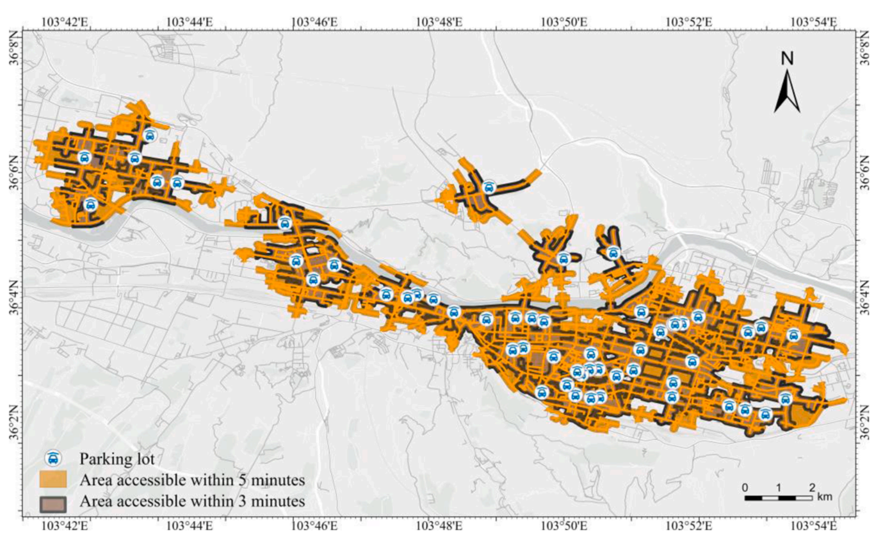

Fig. 5. Spatial graph based on accessibility in the urban district of Lanzhou.

features, the existing hidden features, and spatial information to generate the final hidden state for the current unit. The GA modules, integrated into the gated recurrent unit, enable real-time fusion of multi-view spatial-temporal data. The GA modules capture distinct node relationships, while the gated recurrent unit extracts long-term temporal dependencies.

4. Case study: Lanzhou's urban district

4.1. Data description

Lanzhou, the capital of Gansu Province, is centrally located in China and serves as a crucial hub in the western region as well as a key node in

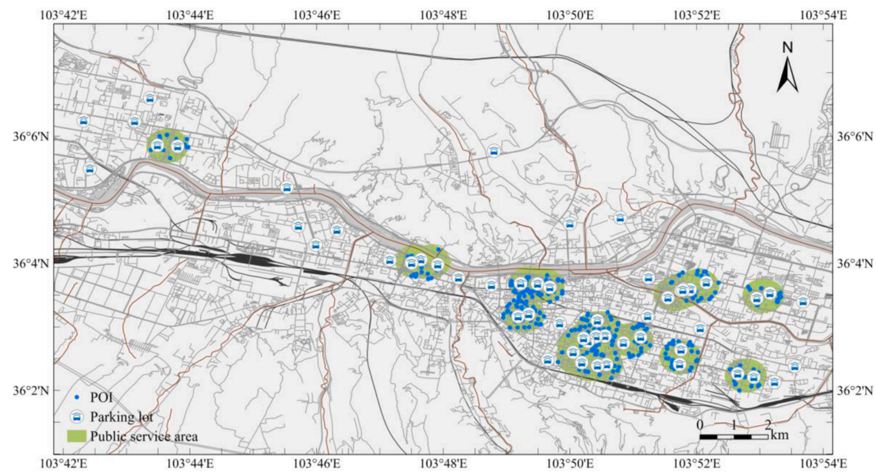

Fig. 6. Spatial graph based on POI in the urban district of Lanzhou.

the Silk Road Economic Belt. As illustrated in Fig. 3, the Yellow River flows through the city of Lanzhou. Despite its importance, the urban area of Lanzhou, as depicted in Fig. 3, exhibits a narrow north-south span and an elongated east-west orientation, limiting parking space availability on the northern and southern peripheries. This spatial configuration further constricts street width, complicating the planning and distribution of parking facilities.

Rapid urbanization has exacerbated existing deficiencies in Lanzhou's urban infrastructure, leading to severe traffic congestion, inadequate road networks, and frequent traffic jams. By the end of 2021, Lanzhou had 1039,200 privately owned vehicles but only 1188 parking lots with a total of 114,500 parking spaces, leading to a parking space ratio of only 0.11 per vehicle. This data was reported in the 2021 Statistical Bulletin on the Development of the Transportation Industry in Lanzhou City, published by the Lanzhou Municipal Transportation Bureau. This imbalance has intensified the city's congestion problems, primarily due to insufficient curb parking provisions. The two small graphs on the right side of Fig. 3 depict two segments of curb parking pay stations in the urban district of Lanzhou, underscoring the pronounced scarcity of curb parking spaces.

To assess the predictive performance of the MGA-GRU model, we employed a dataset from the LTDCG, comprising desensitized parking data from 62 curb parking lots across Lanzhou's urban district. The raw data provided by LTDCG includes the curb parking station ID, total parking spaces, entry time, exit time, parking duration, amount due, amount paid, and payment method. The dataset includes about 9.75 million transactions recorded over 18 months, from July 1, 2019, to December 31, 2020, during the hours of 8:00 AM to 7:40 PM daily. The parking occupancy was calculated by projecting transaction data into hourly intervals, with missing values filled through linear interpolation and normalization to [0,1]. The dataset also contains geographic coordinates for the 62 curb parking lots (Fig. 3). Due to data-sharing restrictions of LTDCG, only sample data are available to readers at https://github.com/linfengqian/Sample-data. Additionally, the POI and road network data were obtained from the Amap Open Platform (https://lbs.amap.com/).

In Fig. 4, each curb parking lot is represented by a blue icon, and the green region depicts the 500-meter radius area surrounding the curb parking lot, as defined in Eq. (3).

Fig. 5 visually illustrates the reachable area of each curb parking lot through the road network, determined by Eq. (5). Curb parking lots are indicated by blue icons, while the surrounding gray area represents the extension accessible within a 3-minute drive at a speed of 20 km/h. Additionally, the yellow area shows the region reachable within a 5-minute drive at the same speed within the road network.

Fig. 6 illustrates the constructed graph based on POI according to Eq. (7). The green area represents the public service region where two adjacent curb parking lots are situated. The blue icons denote the curb parking lots, while the blue dots indicate POI within the vicinity. The POI number in the area signifies the level of mutual influence between the two parking lots. It is important to note that certain curb parking lots may be isolated due to their location in a service area that cannot establish a 500-meter radius connection with others. Consequently, these isolated curb parking lots are less influenced by the POI number in the region, resulting in the absence of interconnected edges between them and other curb parking lots.

After obtaining the three types of multi-view spatial information (α_{ij}^{D}, α_{ij}^{R} and α_{ij}^{P}), they were integrated into a unified spatial graph to facilitate model predictions, utilizing Eq. (8).

4.2. Evaluation metrics

These experiments were implemented using Python 3.10 and PyTorch 1.12 on a desktop computer with Intel (R) Core (TM) i7-13700KF CPU, 32 GB of memory, and one GPU (NVIDIA GeForce RTX 4070 Ti).

The predictive performance of the MGA-GRU model was assessed using five key metrics: R^2, *Acc*, *var*, *MAE*, and *RMSE*. R^2, *Acc*, and *var* are metrics that reflect the model's predictive capabilities, with higher values indicating superior performance. Conversely, *MAE* and *RMSE* are error metrics, where lower values signify greater accuracy in the model's predictions.

(1) Mean Absolute Error (*MAE*)

$$MAE = \frac{1}{MN}\sum_{j=1}^{M}\sum_{i=1}^{N}\left|x_i^j - \widehat{x}_i^j\right| \tag{20}$$

(2) Root Mean Square Error (*RMSE*)

$$RMSE = \sqrt{\frac{1}{MN}\sum_{j=1}^{M}\sum_{i=1}^{N}\left(x_i^j - \widehat{x}_i^j\right)^2} \tag{21}$$

(3) Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum_{j=1}^{M}\sum_{i=1}^{N}\left(x_i^j - \widehat{x}_i^j\right)^2}{\sum_{j=1}^{M}\sum_{i=1}^{N}\left(x_i^j - \overline{X}\right)^2} \tag{22}$$

(4) Accuracy (Acc):

$$Acc = 1 - \frac{||\boldsymbol{X} - \widehat{\boldsymbol{X}}||_F}{||\boldsymbol{X}||_F} \tag{23}$$

(5) Explained Variance Score (var):

$$var = 1 - \frac{Var\{\boldsymbol{X} - \widehat{\boldsymbol{X}}\}}{Var\{\boldsymbol{X}\}} \tag{24}$$

where x_i^j and $\widehat{x}_i^j$ represent the real and predicted values of the curb parking occupancy of parking lot v_i at the prediction time interval j respectively, M and N are the count of parking lots and time samples respectively. $\boldsymbol{X}$ and $\widehat{\boldsymbol{X}}$ denote the sets consisting of x_i^j and $\widehat{x}_i^j$ respectively, and $\overline{X}$ signifies the average of $\boldsymbol{X}$. To measure the prediction error, *RMSE* and *MAE* are employed, where smaller values indicate higher prediction accuracy. The coefficient of determination (R^2), accuracy (Acc), and explained variance score (var) assess the prediction results' ability to represent the actual data, with higher values indicating better prediction performance.

4.3. Benchmark models

To verify the validity of the proposed model, its prediction performance is compared with the following benchmark models in this study.

(1) ARIMA model [7]. The Autoregressive Integrated Moving Average (ARIMA) model fits a parametric model to the observed time series data to predict future parking occupancy.
(2) LSTM model [13]. The Long Short-Term Memory (LSTM) network regulates information transfer using forget gate, update gate, and output gate mechanisms. The use of multiple gates unit in the LSTM model enables longer-term information retention and address the issue of gradient explosion.
(3) GRU model [16]. The Gated Recursive Unit (GRU) model is a simplified alternative to the LSTM model. with update gate and reset gate mechanisms, the GRU effectively controls the information transfer, achieving this with fewer parameters and improved efficiency compared to the LSTM, making it an attractive option for various prediction applications.
(4) T-GCN model [18]. The T-GCN sequentially integrates GCN and GRU to form an integrated model. GCN captures spatial graph based on geographical distance ($\boldsymbol{G}_1$) equally and GRU captures temporal features of parking occupancy data.
(5) LCNN model [24]. The LCNN model combines CNN and LSTM network to efficiently extract spatial and temporal features. CNN extracts spatial features by analyzing data organized in a regular grid structure, while LSTM captures temporal dependencies and extracts temporal features.
(6) GAT-GRU model [39]. The GAT-GRU model utilizes the GAT and GRU for predicting parking occupancy. GAT differentially aggregates spatial graph information based on geographical distance ($\boldsymbol{G}_1$) from neighboring nodes.
(7) MGA-GRU-1G-g: This model performs real-time fusion of spatial-temporal information by embedding the GA gates to the GRU only based on the geographic distance graph ($\boldsymbol{G}_1$).

Table 3
Performance comparison between models.

Time	Model	↓ *MAE*	↓ *RMSE*	↑ R^2	↑ *Acc*	↑ *var*
1 hour	ARIMA	7.6148	9.3619	*	0.7952	0.6259
	GRU	4.5111	7.2638	0.9153	0.8439	0.9223
	LSTM	4.2631	6.5894	0.9302	0.8499	0.9306
	LCNN	4.5727	6.6734	0.9498	0.8695	0.9509
	T-GCN	3.8372	5.5631	0.9503	0.8736	0.9513
	GAT-GRU	3.8093	5.4719	0.9519	0.8757	0.9538
	MGA-GRU-1G-g	3.6141	4.9159	0.9611	0.8840	0.9581
	MGA-GRU	**3.4749**	**4.7158**	**0.9642**	**0.8926**	**0.9643**
2 hours	ARIMA	11.7630	14.3946	*	0.7709	0.5322
	GRU	5.4422	8.5914	0.8815	0.7973	0.8862
	LSTM	4.5317	6.9391	0.9226	0.8268	0.9072
	LCNN	4.9634	7.4170	0.9379	0.8075	0.8923
	T-GCN	4.2386	6.2826	0.9366	0.8036	0.8819
	GAT-GRU	4.2852	6.4030	0.9342	0.7979	0.8767
	MGA-GRU-1G-g	4.0632	5.5431	0.9506	0.8094	0.8871
	MGA-GRU	**3.9084**	**5.3538**	**0.9539**	**0.8670**	**0.9450**
3 hours	ARIMA	13.5630	16.4598	*	0.6406	*
	GRU	5.7209	9.0216	0.8693	0.7949	0.8858
	LSTM	4.7213	7.2804	0.9148	0.8342	0.9152
	LCNN	5.2280	7.9989	0.9277	0.8434	0.9291
	T-GCN	4.4362	6.6324	0.9294	0.8492	0.9317
	GAT-GRU	4.5053	6.8691	0.9242	0.8439	0.9291
	MGA-GRU-1G-g	4.3022	5.9054	0.9440	0.8607	**0.9401**
	MGA-GRU	**4.1642**	**5.7415**	**0.9470**	**0.8585**	0.9377

4.4. Model parameters

The hyperparameters of the MGA-GRU model primarily consist of batch size, learning rate, training epochs, the number of hidden units and the thresholds. In our experiments, we manually set the learning rate to 0.001, the batch size to 64, and the training epoch to 1000. Through parameter tuning, the thresholds for D, R, and B were determined as 500, 5, and 0.3, respectively. To determine the optimal value for the weight decay hyperparameter λ in our model, we conducted a series of experiments. Specifically, we explored a range of λ values and evaluated the model's performance on a separate validation dataset. The experiments were designed to assess the impact of different regularization strengths on the generalization ability of the model. For each λ value, we trained the model using the same architecture and training setup, while monitoring the validation loss and accuracy. By systematically varying λ, we identified the value that resulted in the lowest validation loss, thereby minimizing the risk of overfitting while maintaining model performance. This process enabled us to select a λ value that balances model complexity with predictive accuracy, ensuring robust and generalizable results. Subsequently, we evaluated the model using different numbers of hidden units (8, 16, 32, 64, 100, and 128) and found that 32 yielded the best performance.

To ensure fair comparisons with baselines, we allocated 80 % of the data for the training set (5184 samples) and 20 % for the validation set (1296 samples) in chronological order. Subsequently, we constructed the training and validation sets using a sliding window approach. Specifically, we selected sequences of parking occupancy data with lengths H and F as model inputs and targets, respectively, starting from the initial time point. This window was then incrementally advanced by one time step, repeating the process until reaching the end of the dataset. Importantly, the time durations of the training and validation splits did not overlap. This method provides comprehensive temporal coverage while preserving sequential dependencies, facilitating effective model training and evaluation. The MGA-GRU model was trained using the Adam optimizer, and curb parking occupancy predictions were made for the future 1 hour, 2 hours, and 3 hours.

4.5. Results

4.5.1. Comparison with baseline models

Table 3 demonstrates the predictive performance of the seven benchmark models and the proposed MGA-GRU model across the five-evaluation metrics. The results presented in Table 3 demonstrate that the MGA-GRU model consistently outperforms seven benchmark models in predicting curb parking occupancy for 1, 2, and 3 hours ahead. This

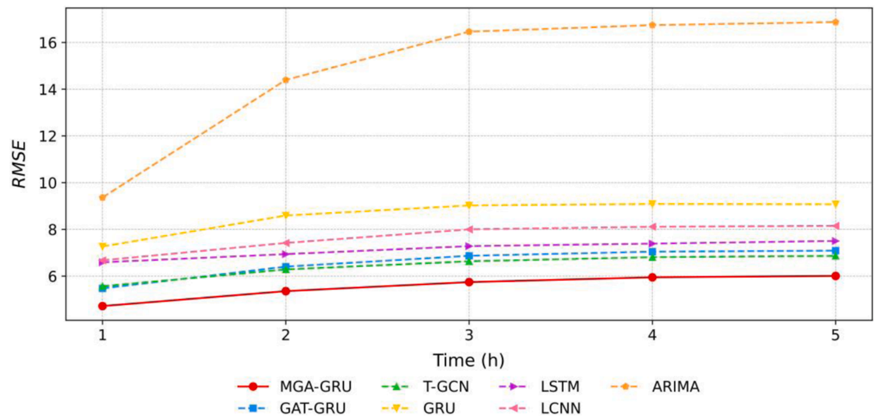

Fig. 7. Trend of *RMSE* for multi-step forecast of each model.

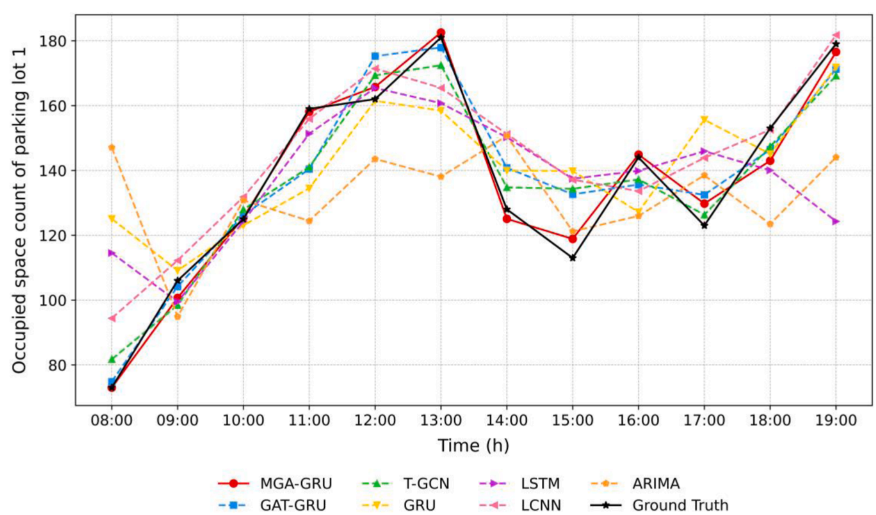

Fig. 8. Visualization of prediction results for curb parking lot 1.

superior performance across multiple time horizons highlights the model's robustness and accuracy, underscoring its effectiveness in providing reliable estimates of curb parking availability. Neural network-based models have superior performance in modeling temporal features compared to traditional ARIMA models based on Table 3. While the LSTM model exhibits slightly superior performance compared to the GRU model in Table 3, attributed to its increased gated units, it is essential to note that the GRU model possesses fewer network parameters and offers faster training speeds. Therefore, the GRU model plays a pivotal role in training complex networks, it is crucial to consider that the GRU model has fewer network parameters and faster training speed. Consequently, both existing composite models (e.g., LCNN, T-GCN, GAT-GRU) and the MGA-GRU model leverage the GRU model or enhance its capability in modeling temporal features.

Upon analyzing the results, our study revealed noteworthy findings. Specifically, the real-time fusion model (GA-GRU-1G-g) exhibited an average reduction in *RMSE* of 12.54 % compared to the non-real-time fusion model (GAT-GRU). Furthermore, leveraging the same embedded graph attention gate for real-time fusion, the multi-view graph model (MGA-GRU) demonstrated an average reduction in *RMSE* of 3.42 % compared to the single-view graph model (GA-GRU-1G-g). These findings underscore the efficacy of real-time fusion techniques and the added benefit of employing multi-view graph structures in enhancing prediction accuracy in the curb parking occupancy prediction.

In the realm of spatial-temporal feature modeling, fusion models that integrate spatial-temporal features (T-GCN, GAT-GRU, and MGA-GRU) have shown superior performance in multi-scale prediction across all evaluation metrics, surpassing temporal feature models (ARIMA, LSTM, and GRU). This finding emphasizes the effectiveness of incorporating spatial features to enhance the predictive. While the LCNN model outperforms the GRU model in multiple time scales of prediction, it only exhibits superiority in specific metrics (R^2, *Acc*, and *var*) compared to the LSTM model. This performance difference could be attributed to the limited efficiency of the CNN module employed in the LCNN model.

4.5.2. Multi-step forecast analysis

To evaluate the long-term prediction capability of the model, a multi-step prediction approach is employed using a rolling window. Fig. 7 depicts the *RMSE* fluctuation across various models and prediction steps.

Based on the findings, it is evident that as the prediction step size increases, the ARIMA model exhibits the most significant overall error and the largest variation compared to other models. Among the deep learning models, the LSTM model demonstrates a noticeable advantage in long-term prediction with the smallest error fluctuation due to its higher number of gated units. However, the LSTM model still displays a relatively high overall error. The error curves of the spatial-temporal models (LCNN, T-GCN, GAT-GRU, and MGA-GRU) exhibit smoother changes compared to the GRU model, indicating that the inclusion of spatial features enhances the long-term prediction capability of the model.

As the prediction step size increases, the MGA-GRU model

Table 4
Performance comparison between models of ablation experiment.

Time	Model	↓ *MAE*	↓ *RMSE*	↑ R^2	↑ *Acc*	↑ *var*
1 hour	GAT-GRU	3.8093	5.4719	0.9519	0.8757	0.9538
	MGA-GRU-1G-g	3.6141	4.9159	0.9611	0.8840	0.9581
	MGA-GRU-1G-a	3.6230	4.9338	0.9609	0.8792	0.9546
	MGA-GRU-1G-p	3.6490	4.9600	0.9609	0.8838	0.9580
	MGA-GRU-2G-ga	3.5951	4.8940	0.9615	0.8760	0.9520
	MGA-GRU-2G-pa	3.5907	4.8620	0.9619	0.8673	0.9450
	MGA-GRU-2G-gp	3.5290	4.8400	0.9620	0.8800	0.9500
	MGA-GRU	**3.4749**	**4.7158**	**0.9642**	**0.8926**	**0.9643**
2 hours	GAT-GRU	4.2852	6.4030	0.9342	0.7979	0.8767
	MGA-GRU-1G-g	4.0632	5.5431	0.9506	0.8094	0.8871
	MGA-GRU-1G-a	4.0677	5.5558	0.9504	0.8018	0.8770
	MGA-GRU-1G-p	4.1320	5.6620	0.9480	0.8092	0.8860
	MGA-GRU-2G-ga	4.1073	5.6246	0.9491	0.8040	0.8810
	MGA-GRU-2G-pa	4.0430	5.5120	0.9511	0.7985	0.8736
	MGA-GRU-2G-gp	3.9220	5.4320	0.9526	0.8097	0.8870
	MGA-GRU	**3.9084**	**5.3538**	**0.9539**	**0.8670**	**0.9450**
3 hours	GAT-GRU	4.5053	6.8691	0.9242	0.8439	0.9291
	MGA-GRU-1G-g	4.3022	5.9054	0.9440	0.8607	0.9401
	MGA-GRU-1G-a	4.2762	5.8730	0.9446	0.8410	0.9210
	MGA-GRU-1G-p	4.3500	6.0090	0.9420	0.8590	0.9380
	MGA-GRU-2G-ga	4.4260	6.1030	0.9402	0.8536	0.9330
	MGA-GRU-2G-pa	4.2930	5.8690	0.9446	0.8394	0.9190
	MGA-GRU-2G-gp	**4.0930**	**5.6983**	**0.9478**	**0.8650**	**0.9438**
	MGA-GRU	4.1642	5.7415	0.9470	0.8585	0.9377

consistently demonstrates reduced overall errors compared to other models. Its error curve consistently maintains the lowest position and exhibits a smoother trend. Notably, the fifth-step prediction error is only 1.29 units higher than that of the initial step, surpassing the performance of alternative spatial-temporal prediction models. These findings underscore the superior long-term predictive capacity of the MGA-GRU model.

4.5.3. Prediction results visualization

To provide a comprehensive illustration of the predictive capabilities of each model, we selected a random curb parking lot from the test set and applied a rolling window approach to forecast the occupied parking space numbers over the course of one day. Fig. 8 showcases the prediction results of each model.

The analysis of the prediction results unveils distinctive characteristics among the various models. The ARIMA model demonstrates a limited ability to capture local maxima and minima despite adequately simulating the overall trend. Similarly, temporal feature models such as GRU and LSTM exhibit insufficient predictive capacity for the local extremities of the curve, despite their relatively accurate trend simulations. In contrast, spatial-temporal feature models, such as LCNN, T-GCN, and GAT-GRU, demonstrate enhanced prediction capabilities for local maxima and minima. Nevertheless, their performance remains subpar compared to the MGA-GRU model proposed in this study. This gap could be attributed to the spatial information lag and insufficient spatial-temporal information fusion in existing spatial-temporal models.

4.6. Ablation study

A series of ablation experiments were conducted to comprehensively evaluate the contributions of the MGA-GRU network model and its components. The MGA-GRU model introduces multi-view graph construction and real-time fusion of the GA gates as key innovations. To validate the efficacy of these enhancements, the following additional experiments were sequentially conducted:

(1) MGA-GRU-1G-g: Real-time fusion of spatial-temporal information through GA gates based on the geographic distance graph ($\boldsymbol{G}_1$).
(2) MGA-GRU-1G-a: Real-time fusion of spatial-temporal information through GA gates based on the accessibility graph ($\boldsymbol{G}_2$).
(3) MGA-GRU-1G-p: Real-time fusion of spatial-temporal information through GA gates based on the POI graph ($\boldsymbol{G}_3$).
(4) MGA-GRU-2G-ga: Real-time fusion of spatial-temporal through GA gates based on both the geographic distance graph ($\boldsymbol{G}_1$) and accessibility graph ($\boldsymbol{G}_2$).
(5) MGA-GRU-2G-pa: Real-time fusion of spatial-temporal information through GA gates based on both the POI graph ($\boldsymbol{G}_3$) and accessibility graph ($\boldsymbol{G}_2$).
(6) MGA-GRU-2G-gp: Real-time fusion of spatial-temporal information through GA gates based on both the geographical distance graph ($\boldsymbol{G}_1$) and POI graph ($\boldsymbol{G}_3$).

Table 4 presents the predictive performance of six models from the ablation experiments, the GAT-GRU model, and the proposed MGA-GRU model across the five-evaluation metrics. The MGA-GRU-2G-gp and MGA-GRU models demonstrated the best performance based on Table 4. The findings indicate that removing graphs from the MGA-GRU model decreases predictive performance, highlighting the critical role of multi-view graphs in capturing spatial relationships. Notably, there is a significant decline in model performance when the entire graph is eliminated, emphasizing the essential nature of spatial graph features.

To validate the contribution of each component within our model, we conducted a series of ablation experiments. These experiments systematically removed individual components, allowing us to quantify their impact on the model's performance. The results confirm that each component—whether the GA gates, the multi-view graph structure, or the real-time fusion mechanism—significantly contributes to the overall predictive capability of the model. These ablation studies not only reinforce the effectiveness of our approach but also provide insights into the relative importance of each component in enhancing prediction accuracy.

4.7. Discussion

Existing studies on curb parking occupancy prediction have yielded valuable insights but also present notable limitations. First, while spatial-temporal feature fusion is essential, the inherent complexity in combining these features often hinders accuracy. Additionally, most existing methods focus predominantly on geographic proximity between parking lots, disregarding other influential factors such as accessibility and surrounding infrastructure.

To address these gaps, our study introduces advanced real-time fusion techniques designed to improve predictive accuracy in curb parking models. Specifically, the proposed real-time fusion model (GA-GRU-1G-g) outperforms a non-real-time counterpart (GAT-GRU), achieving a 12.54 % reduction in prediction error. This reduction highlights the benefit of real-time fusion in effectively capturing spatial-temporal dependencies and enhancing model precision.

Furthermore, this research explores multi-view graph structures in curb parking prediction. The multi-view graph model (MGA-GRU), which incorporates diverse spatial factors beyond simple geographic distance, achieves a 3.42 % reduction in average prediction error relative to the single-view model (GA-GRU-1G-g). These results suggest that integrating varied spatial influences enhances both accuracy and reliability in occupancy forecasting.

Despite its strengths, the model's complexity results in extended training time. However, this added computational cost minimally impacts prediction timeliness, maintaining a delay within a few seconds.

To optimize efficiency, we recommend scheduling model training to align with system workflows, thus balancing high accuracy with real-time applicability.

Overall, this study underscores the importance of real-time fusion and multi-view graph approaches in curb parking prediction, offering actionable insights for improving urban parking management, reducing congestion, and optimizing resource allocation.

5. Conclusions and future work

As cities expand and congestion rises, managing curb parking becomes crucial for urban planning and enhancing residents' quality of life. Predicting curb parking occupancy with precision is essential for resource optimization and waste reduction. Key findings from this study include:

(1) **Real-time spatial-temporal fusion**: The MGA-GRU model, which integrates two GA gates within a gated recurrent unit, enables real-time fusion of spatial and temporal information. This integration improves the scientific and logical structure of curb parking occupancy prediction models by allowing for differential aggregation of spatial data from neighboring nodes, resulting in more accurate and reliable predictions.
(2) **Multi-view spatial graph construction**: The construction of a multi-view spatial graph that incorporates geographic distance, accessibility, and POI data, further improves the model's ability to capture complex spatial relationships influencing parking occupancy.
(3) **Enhanced predictive performance**: This study validates the MGA-GRU model's predictive strength in curb parking occupancy through a case study in central Lanzhou. The results show that the MGA-GRU model surpasses baseline methods, achieving higher predictive accuracy. This improved accuracy enables more effective planning and real-time management of curb parking systems, enhancing urban mobility and parking accessibility in densely populated areas.

While the MGA-GRU model represents a significant advancement in curb parking occupancy prediction, there are several avenues for future research that could further enhance its capabilities and applicability. In the future, this research can further expand from the following perspectives:

(1) **Optimize for real-time application**: The model's high training time poses a challenge for real-time deployment. Future research should focus on enhancing computational efficiency through model pruning or quantization to reduce training time without affecting accuracy.
(2) **Incorporate diverse data sources**: Adding real-time traffic, weather conditions, or social event data can further improve accuracy and robustness, offering a more holistic view of factors influencing curb parking occupancy and enhancing predictive capabilities.
(3) **Broader regional generalization**: Expanding the model's application to cities with varied traffic and accessibility patterns would enable a more thorough evaluation of its adaptability to different urban environments, addressing the current limitation of testing solely in Lanzhou. This would allow for a more complete understanding of the model's potential across diverse settings.

CRediT authorship contribution statement

Chonghui Qian: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Kexu Yang**: Writing – original draft, Visualization, Validation, Software, Methodology, Data curation. **Jiangping He**: Writing – review & editing. **Xiaojing Peng**: Writing – review & editing. **Hengjun Huang**: Writing – review & editing, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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